Kabir Singh

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Academic Education

Albert Dorman Honors College (ADHC) & Jordan Hu College of Sciences & Liberal Arts (JHCSLA), New Jersey Institute of Technology, Newark, New Jersey, USA — Sep '23 – May '27 (Expected)

- Bachelor of Science; Major – **Mathematical Biology**
- Minors – **Biomedical Engineering in Technology, Philosophy, History**
- ADHC Honors Tracks – **Medical Humanities, Global Studies**
- *Academic Excellence, Honors Merit, Honors Residential Scholarship Recipient (Tuition + Board)*

Faculty of Engineering, Lunds Tekniska Högskola, **Lunds Universitet**, Lund, Sweden — Jan '25 – Jun '25

- Fellow, Institute of International Education's Global Engineering Education Exchange (Global E3)

Research

◇ *Computational and Mathematical Biology* ◇

Undergraduate Researcher, **Computational Biology Team Lead** at KumarLab. PI: Dr. Vivek Kumar, University of Houston (Prev. NJIT) — Oct '23 – Present

- Lead undergraduate and graduate students on designing and developing peptide-based therapeutics
- Started at NJIT and continued computational work with lab relocation to Houston
- Taught 15+ associates in lab
- Received > $15,000 in direct funding for projects

Undergraduate Researcher, Computational Bionanotechnology. PI: Dr. Stefano Angioletti-Uberti, Imperial College London — Jun '26 – Present

- **BAGEL: Biomolecular Algorithm for Guidance in Energy Landscape**
- Computational modelling of soft-matter and biomaterial systems
- Applying statistical mechanics, molecular dynamics, and machine-learning methods to problems in enzyme design and broader protein design

Undergraduate Researcher, Mathematical Neuroscience. PI: Dr. Horacio Rotstein, NJIT — Sep '26 – Present

- Deriving reduced mathematical descriptions of biophysical neuronal models to characterize their dynamic properties

Undergraduate Researcher, **Mathematical Modeling**. PI: Dr. James Maclaurin, NJIT — Oct '25 – May '26

- Applying Hawkes self-exciting point process to model secondary nucleation dynamics in amyloid formation

◇ *Intellectual and Cultural History* ◇

Emerging Scholars Research Fellow, Early Modern Sikh History. Mentor: Satnam Singh, Harvard Sikh Center — Sep '26 – Present

- *Dissemination and the making of Khalsa authority in a contested panth*: examining how the Khalsa asserted dominance over the wider Sikh world within decades of its founding, against rival claimants including the Bandai Khalsa, Minas, and Ram Raiyas
- Arguing that the consolidation of "authentic Sikh teachings" through katha was instrumental to that dominance; working from Gurmukhi and Persian primary sources, including works attributed to Bhai Mani Singh, hukamname, and rahit-nama literature

Undergraduate Researcher, **History of Medicine & Bioethics**. PI: Dr. Stephen Pemberton, NJIT — Sep '26 – Present

- Synthesizing legal, clinical, and cultural developments in American end-of-life care from the late twentieth century to the present
- Developing original scholarship for submission to a peer-reviewed humanities journal

Undergraduate Researcher, **Digital History & Urban Studies**. PI: Dr. Louis Hamilton, NJIT — Jan '24 – May '24, Jan '26 – May '26

- Digital mapping of how religious institutions shaped urban form in medieval and early modern Italy; field research conducted in Rome and Florence
- *Power vs. the People* (HIST 328; 18 sites): layered Counter-Reformation Marian shrines against Julius II's 1508 papal axis on the Via Giulia
- *Processions and Power* (HIST 329; 22 sites): traced how Florence's expansion turned liturgical processions into contested displays of ecclesiastical and household prestige

◇ *Emergency Care and Health Services* ◇

Undergraduate Researcher, Collegiate Emergency Medical Services. Dr. Chang Yaramothu, NJIT — Sep '25 – Present

- Studied the resources of ∼50 US collegiate EMS squads, examining their organizational setups and capabilities in relation to their effectiveness

FUNDING

Undergraduate Research Opportunities Programme (UROP), Imperial College London — Jun '26 – Jul '26

- Competitive research bursary awarded through the Department of Materials, Faculty of Engineering
- Project: Using protein language models to mimic natural ligands. Supervisor: Prof. Stefano Angioletti-Uberti
- Personal compensation included

NSF I-Corps Regional Program, Northeast Hub — May '26 – Jun '26

- Co-Founder & Technical Lead, Axioma Therapeutics
- Selected into competitive NSF cohort; led customer discovery and lean-startup validation for cyclic peptide therapeutics targeting degenerative eye diseases
- Project funding awarded to support customer-discovery activities

Grace Hopper AI Research Institute, Computational Protein Design — May '25 – Aug '25

- NJIT grant to develop a course on computational protein design
- Teaching Assistant

The Honors College Moonshot Grants for Engaged Scholarship — Jun '25 – Jul '25

- Endowed award to develop interdisciplinary projects that address important social and cultural questions
- Designing Male Contraceptive
- Award providing labs with funds to support undergraduate-led projects
- Personal compensation included. One recipient per annum

Male Contraceptive Initiative (MCI) — Jan '25 – Jun '25

- Award by the Youth Advisory Board providing funding to support grantee-led project
- Designing Male Contraceptive
- Personal compensation included

NJIT Undergraduate Research & Innovation (URI) — Oct '24 – May '25

- Student Seed Grant award for a Phase-2 longitudinal project on application-based research funding lab to support undergraduate-led project
- Developing cyclic peptide therapeutic for Multiple Sclerosis

Travel Award – National Contraception Meeting — May '25

- Awarded by the Male Contraceptive Initiative for travel to Durham, NC, supporting a conference presentation

NJIT Honors Summer Research Institute (HSRI) — Jun '24 – Jul '24

- Grant award by ADHC and URI providing funds to support undergraduate-led projects
- A combination of Multiple Sclerosis project and Male Contraceptive project
- Personal compensation included

PUBLICATIONS

Peer-Reviewed

– C. Heffernan, O. West, **K. Singh**, J. Dodd-o, M. James, A. Baradi, K. Rybak, A. Lateef, S. Elkabes, V. Kumar. A novel cyclic peptide that blocks human autoimmune antibodies for the treatment of multiple sclerosis. *European Journal of Medicinal Chemistry*, 119173 (2026). doi:10.1016/j.ejmech.2026.119173

Manuscripts in Preparation

– **K. Singh**, C. Heffernan, O. West, J. Dodd-o, V. Kumar. Machine learning for rational design of a ***** targeting cyclic peptide-based male contraceptive. Publication [in preparation], **Patent application** [in preparation], 2026
– J. Lau, D. Yun, **K. Singh**, A. Patananan, N. Jadavji. Academic Publishing as a Launchpad: How Publishing Research in an Undergraduate Journal Shapes Future Careers. Publication [in preparation], 2026

PRESENTATIONS

M. Kotta, G. Victorio, G. Karmacharya, **K. Singh**, N. Giovannetti, B. Shadpoor, J. Dodd-O, V. Kumar. Putative computational design of novel inhibitors to modulate de-novo targets for CB1 receptor.

UH (Undergraduate Research Day)	Poster + Podium	2026

K. Singh, C. Heffernan, J. Dodd-O, V. Kumar. Development of a novel male contraceptive using cyclic peptides to inhibit fertilization.

AAP/ASCI/APSA Joint Meeting	Poster	2026
NJIT AI Exploration Day	Poster	2026
NJIT ADHC (Honors Interdisciplinary Research Forum) **First Place**	Poster	Fall 2025
NCM (National Contraception Meeting)	Poster	2025
NJIT ADHC (Honors Interdisciplinary Research Forum)	Poster	Spring 2025
NJIT (Dana Knox Research Showcase)	Poster	2025
NJIT (Undergraduate Research & Innovation)	Abstract + Podium	2025
BMES (Biomedical Engineering Society) *[Co-Presenter]*	Abstract + Poster	2024
NJIT (Undergraduate Research & Innovation)	Abstract + Podium	2024

K. Singh, J. Lau, H. Gerber, A. Alibilli, L. Goralsky. Mitigating Knowledge Drain: A Case Study in Optimizing Workflow and Training in a High-Turnover, International Peer-Reviewed Undergraduate Science Academic Journal.

NCUR	Poster	2026

ACADEMIC SERVICE

◇ *Editorial and Peer Review* ◇

22nd Editor-in-Chief, Associate Editor (Former), Journal of Young Investigators — Dec '24 – Present

– Manage timely publication of high-quality scientific content
– Develop initiatives to further the journal's mission
– Former specialization in Biology, Physical Sciences, and Social Science Departments

Grant Reviewer, MCI YAB Undergraduate Project Grant — Dec '25 – Jan '26

◇ *Teaching and Mentorship* ◇

Research Lead, Governor's STEM Scholars, Research & Development Council of New Jersey — Sep '26 – May '27 (Expected)

– Guiding a team of high school Scholars through a year-long research and engineering design project anchored to a UN Sustainable Development Goal and addressing a New Jersey community need
– Overseeing team meetings, deliverables, timeline, and budget, and preparing the team's final presentation across four statewide conferences on STEM in government, academia, and industry
– Appointment within a public–private partnership of the Office of the Governor, NJ Department of Education, and NJ Office of the Secretary of Higher Education; $1,000 completion stipend

Teaching Assistant, Honors First Year Seminar — Sep '24 – Dec '24, Sep '25 – Dec '25

Teaching Assistant, Grace Hopper AI Research Institute — May '25 – Aug '25

– Computational Protein Design

◇ *National and Professional Service* ◇

Research Chair, American Association of Physicians of Indian Origin (AAPI) PreMed Board — Jun '25 – Present

– Lead committee and organize events to achieve research-related objectives on a national level to support students, researchers, physicians, and more of an Indian origin

Board Member, MCI Youth Advisory Board (YAB) — Aug '25 – Present

Member, Pre-Medical Ad Hoc Committee, APSA — Mar '26 – Present

Member, Goldwater Ambassadors Program — Aug '25 – Present

– Selected into competitive process as a mentee, mentored by Goldwater Scholars
– Program specially designed for students from underrepresented or underserved backgrounds

Member, ARTISAN Program — Apr '25 – Present

– Inducted in the NSF ART Integrated Student Ambassador Network (ARTISAN) program, based off commitment to translational research in my work

◇ *University Service* ◇

Member, Student Conduct Board — Dec '24 – Present

– Serve as adjudicator on academic and behavioral cases in student conduct hearings, along with other members of the board, including faculty and NJIT administration

Chair, Chair-in-Training (Former), Highlander Integrity Council — Sep '24 – Present

Member, Inclusive Excellence Student Advisory Board — Feb '25 – Present

Member, Multifaith Council — Jan '25 – Present

Organizer, 'It's Different for Girls' Documentary Screening & Panel, NJIT ADHC & CSLA Co-Sponsored Colloquium — Mar '26

NJIT Representative, BioNJ 33rd Annual Dinner Meeting & Innovation Celebration — Feb '26

Steering Committee Lead, ADHC First Year Retreat — Mar '25 – Aug '25

Advisor, President (Former), Secretary (Former), Honors Hall Council — Feb '24 – May '25

ADHC Representative, National Collegiate Honors Council Annual Conference — Nov '24

Lectures & Features

– Invited Speaker (forthcoming). Student roundtable with Margo Chaly, New Jersey's Secretary of Education. NJIT, 2026
– Invited Speaker. Presentation on interdisciplinary research in medicine and healthcare to NJIT leadership, including the Interim President and Interim Provost. University Advancement Dinner, NJIT, 2026
– Lecture Series. Ethics in Science and Research. Science Mentorship Institute, 2026
– Session Moderator. Student Oral Presentation (Communications). NCUR, 2026
– Featured Presenter. 'Navigating peer review and the student journal landscape'. "Shaping Asia's Future: Development, Innovation, and Society in Transition" organized by Nazarbayev University, Asian Universities Alliance, 2026
– Featured Panelist. 'It's Different for Girls' Documentary Screening & Panel. NJIT ADHC & CSLA Co-Sponsored Colloquium, NJIT 2026
– NJIT Makes: Undergraduate Research & Innovation, NJIT Promotional Material. NJIT, 2026
– Invited Speaker. Presentations to the NJIT Board of Trustees and the President's Town Hall with squad leadership, at the request of NJIT President Teik Lim. NJIT First Aid Squad, 2025
– Albert Dorman Honors College, Honors in Action 2025 (Research, Award, and Study Abroad Experience). ADHC Publication. NJIT, 2025
– Nayib Morán. '150+ Honors Scholars Showcase Findings at Interdisciplinary Research Forum'. Article Feature. NJIT News, 2025
– Featured Speaker. ADHC Advising Day. NJIT, 2024 & 2025
– Albert Dorman Honors College, Honors in Action 2024 (Research and Study Abroad Experience). ADHC Publication. NJIT, 2024
– Gurnoor Kaur. 'Undergraduate Student Researchers on Campus'. Article Feature. NJIT (Student Newspaper – The Vector), 2024
– K. Singh. 'Introduction to PyMOL and Linear Peptide Design'. Lecture. NJIT (Genetically Engineered Machines), 2024

Certifications & Licenses

- New Jersey EMT License
- ACS "Stop the Bleed" Certification & Instructor License
- OSHA General Industry Safety & Health Certification

- CITI Biomedical Research Certification
- CITI RCR Basic Certification
- American Heart Association BLS & CPR for Adult/Child/Infant Certification

Leadership

Ethics Coordinator, Mentor (Neuro + Bio), Ethics Curriculum Dev (Former), Admission Officer, Symposium Committee Organizer, Grant Writer, Science Mentorship Institute (Sci-Mi)	May '24 – Present
Public Relations Chair, EMT, NJIT First Aid Squad	Oct '25 – Present, Jan '24 – Present
Khalsa School Virsa (Sikh History) Teacher, Sikh Sabha of New Jersey	Sep '20 – Present
Youth Camp Organizer & Coordinator, Sikh Sabha of New Jersey	Aug '21 – Present
Content Manager, The Mardana Project	May '26 – Present
Stop the Bleed Instructor, NJIT	Oct '25 – Present
Vice President, President (Former), Events Coordinator (Former), Sikh Association of NJIT (SANJIT)	Dec '23 – Present
Stop the Bleed Instructor (Hybrid English–Punjabi), NJ Gurdwaras, NJ Office of Homeland Security & Preparedness (NJOHSP) & Sikh Youth Alliance	Jan '26 – May '26
Youth Camp Counselor, Guru Nanak Sikh Heritage	Aug '21 – Aug '25
President, Treasurer (Former), Co-Founder, NJIT MedicZero	Nov '23 – May '26
Co-Founder, Delta Epsilon Mu (Pre-medical Professional Fraternity)	Feb '24 – May '25
Fellow, Honors Summer Research Institute Fellowship	Jun '24 – Jul '24, Jun '25 – Jul '25
EMT Senior Squad Member, Cadet Crew Chief (Former), Marlboro First Aid Squad	Apr '22 – Jan '25
Fellow, No Patient Left Behind Fellowship	Jun '24 – Aug '24
Young Scholar, American College of Cardiology	May '23 – Apr '24

References and letters of recommendation available upon request.